

8. Given an array of N integers, you are asked to find out whether the given array is an arithmetic series — which is a sequence of numbers in which there is a constant difference between any two consecutive numbers. What is the complexity of your function? You are then given a modified problem for which your task is to convert a given array into an arithmetic sequence by dropping off those array elements which prevent it from being an arithmetic sequence. How will you solve this problem? What is the time complexity of your function?
 9. You are given a list of N integers, for which the numbers can only be positive. You are now being asked to write code to create two lists of size $N/2$ such that the sum of the elements of the two sub-lists is equal. Is it possible to create such sub-lists? Your function should return the sum of the sub-list elements if it is possible to split the original list into two such sub-lists or return 0 if the former is not possible. What is the time complexity of your function? (Note that you can move the elements of the list.) You are now told that the original list can either contain positive or negative integers, and are expected to solve the above problem. What are the changes you will make to your earlier solution?
 10. Consider the problem given in (9). Now, instead of splitting the list into two sub-lists of equal sizes such that the elements of the two sub-lists are equal, you are asked to split the list at any arbitrary index such that the difference of the sum of the two sub-lists is minimum among all possible splits of the original list. Unlike Problem 10, you cannot move around the elements of the list. What is the time and space complexity of your function?
 11. You are given two sentences, S_1 and S_2 , each containing N and M words, respectively. You are now asked to convert S_1 into S_2 by either deleting a word or adding a new word. (Modifying word1 to word2 can be considered as word deletion, followed by word addition.) Each insert operation has a cost of +2 and each delete operation has a cost of +1. Given two arbitrary sentences S_1 and S_2 , the edit distance of (S_1, S_2) is the minimum total cost of operations needed to convert S_1 into S_2 . You are asked to write a function to compute the edit distance. What is the time and space complexity of your function?
 12. You are given a data set containing the weekly rainfall experienced in Bengaluru for the last 20 years. The data contains the week dates, and the corresponding rainfall in millimetres, for that week. Your first task is to find out what the outliers in your data are. How will you do this?
 13. Given a classifier, can you define the terms ‘sensitivity’ and ‘specificity’ of your system? Let us assume that you have built a classifier which can predict whether a tumour is benign or malignant, based on certain tumour characteristics. Your intention is to build a classifier that can detect all true malignant tumours, even if some of the benign tumours get falsely reported as malignant. Which of the two metrics’ sensitivity or specificity should you try to improve in this case? Now assume that you have been asked to use a classifier to predict whether a student will pass the examination or not. Your system should be designed such that it does not report a failing student as passing even if it mispredicts certain passing students as potential failures. In that case, which of the two metrics’ sensitivity or specificity should you try to improve?
 14. Different activation functions can be employed for the neurons such as rectilinear, tan-h, sigmoid, etc. Can you explain the factors involved in the choice of activation functions? Are there any criteria that a function needs to satisfy for it to be considered as an activation function in a neural network trained using back propagation?
 15. You have built a text classification system for classifying news articles, as we discussed in Question 2 earlier. Now you are asked to write an application for classifying the subject of tweets/micro-texts. Will you be able to use the same application? If yes, why? If not, give your reason.
- I would like to close this month’s column by wishing our readers a great year ahead. Many of us make new career resolutions at the beginning of the year. I have received a number of emails from our readers asking how they can become proficient in data science, AI, machine learning, RPA (robotic process automation), etc. There are a couple of points I wanted to share with our readers in this context.
- When you start exploring a new area of computer science, be it machine learning, block chain, RPA or anything else, everything initially seems complex and difficult. You need to recall how the areas you are now pretty strong in felt to you when you were exploring them for the first time. Please don’t feel daunted by the new areas you need to develop skills for. A growth mindset is more important than any skills or knowledge you may pick up, because skills can become obsolete as technology evolves. But a growth mindset will allow you to become knowledgeable in new areas as you progress in your career.
 - Also, do remember the power of compounding. Whichever area of computer science you choose to become proficient in for this year, please invest in a daily reading and coding schedule on topics in that area. The key is to adhere to the daily discipline even if you don’t see any tangible results in the beginning. Make sure to share your newly acquired knowledge by creating blog posts or questions-and-answers.
- Feel free to reach out to me over LinkedIn or email if you need any help in your new study journey. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! 

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